

YZV 447E TERM PROJECT REPORT
Fall 2025
Report 2: Implementation
Due: 21/12/2025

Experiment summary

- **Hardware:** MacBook
- **Software:** PsychoPy installed under MacOS
- **Stimuli:** Cropped face images from the KDEF dataset (controlled by an Excel condition file).
- **Design revisions:**
 - **Cropping:** Based on the instructor's feedback to prevent the ceiling effect, we cropped the original KDEF images to oval/rectangular shapes to remove hair and ear cues. This forces the participant to focus on internal facial features.
 - **Duration:** Originally, we considered a fixed duration (1000 ms). However, to provide participants enough time to process the cropped faces which are harder to recognize, we revised the design to keep the stimuli on the screen **until a button is pressed**.
 - **Response Keys:** We updated the response mapping to be more intuitive: **'F'** for Female and **'M'** for Male.

Programming information

- **Input:** Human face photographs selected from the KDEF (Karolinska Directed Emotional Faces) dataset.
 - **Selection Criteria:** To maintain a balanced design with 120 trials, we selected specific IDs:
 - **Females:** IDs 15 to 35 (excluding ID 28 due to missing data).
 - **Males:** IDs 01 to 20.
- **Stimulus types:** 2 types: **Male face, Female face**.
- **Stimulus categories:** 3 categories of Emotion: **Happy, Angry, Neutral**.
- **Stimulus order: Pseudo-randomized:** The stimuli were presented in a randomized order using the PsychoPy randomizer. To ensure consistency across all participants, a fixed **random seed** (seed=42) was set in the experiment loop. This ensured that while the sequence appeared random to the participant, every subject encountered the exact same stimulus order.
- **Stimulus manipulation:**
 - **Fixation:** A fixation cross appears for **0.5 seconds** before each face.
 - **Variables:** The independent variables defined in the condition file are "Gender" (Male/Female) and "Emotion" (Happy/Angry/Neutral).

Usage instructions

- **Usage:** Open the PsychoPy project (gender_task.psyexp), click on the **"Run"** button, and enter the participant's name/ID in the dialog box.
- **Data collection procedure:** Before the experiment starts, a dialog box appears on the screen requesting demographic information. The participant enters the following data, which is automatically logged into the output file:
 - **Age:** The participant's age.
 - **Gender:** The participant's gender.
 - **Education:** Calculated in years (K-12 is counted as 12 years; each completed year at the university adds one year).
 - **Handedness:** The participant's dominant hand (Right/Left) or the hand used to press the buttons.
- **Instructions to the subject:** The following text is presented on the initial screen:

"You are about to see a series of human faces on the screen. Your task is to decide the gender of the person, **IGNORING** the facial expression (angry, happy, or neutral). If the person you see is **FEMALE**, press the **F** key. If the person you see is **MALE**, press the **M** key. Please try to respond as quickly and accurately as possible. Press the **spacebar** to begin when you are ready."

Output file(s)

- **Name of attached code file:** gender_task.psyexp
- **Name of attached stimulus file:** conditions.xlsx
- **Name of example output file:** test7_gender_task.csv

Endorsement of team members

- | | |
|--|--|
| 1. Yusuf Oğuz - 150220322 | SIGN:  |
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APPENDIX

Stimulus file Contents

The following table represents the structure of our conditions.xlsx file (showing the first 12 rows as an example):

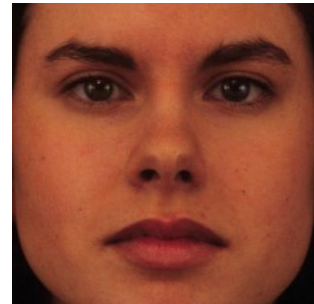
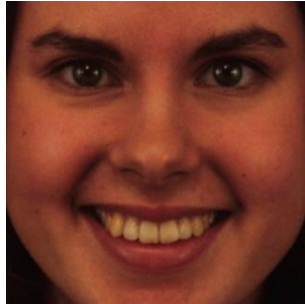
ImageFile	Gender	Emotion	CorrectAns
stimuli/F15	Female	Angry	f
stimuli/F15	Female	Happy	f
stimuli/F15	Female	Neutral	f
stimuli/F16	Female	Angry	f
stimuli/F16	Female	Happy	f
stimuli/F16	Female	Neutral	f
stimuli/F17	Female	Angry	f
stimuli/F17	Female	Happy	f
stimuli/F17	Female	Neutral	f
stimuli/F18	Female	Angry	f
stimuli/F18	Female	Happy	f
stimuli/F18	Female	Neutral	f

Output file Contents:

The following table shows the raw data output from a pilot run (showing relevant columns):

ImageFile	Gender	Emotion	key_resp.keys	key_resp.rt	key_resp.corr
stimuli/F35HA.jpg	Female	Happy	f	2.740	1
stimuli/F17NE.jpg	Female	Neutral	f	0.677	1
stimuli/M18HA.jpg	Male	Happy	f	0.492	0
stimuli/M09HA.jpg	Male	Happy	m	0.527	1
stimuli/M05HA.jpg	Male	Happy	m	0.505	1
stimuli/F15AN.jpg	Female	Angry	m	0.669	0
stimuli/F31NE.jpg	Female	Neutral	f	0.536	1
stimuli/M09AN.jpg	Male	Angry	m	0.456	1
stimuli/F25AN.jpg	Female	Angry	f	0.612	1
stimuli/M13NE.jpg	Male	Neutral	f	0.466	0

Example Stimuli:



As seen above, hair and non-facial features were removed to focus attention on facial expressions.